Kolawole (Paul) Oshikoya - Technical Project Portfolio

Focus Areas: Cloud Data Platforms, AWS Infrastructure, Data Governance, DevOps & CI/CD
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1. Cloud Data Platform & EMR Cluster Lifecycle Management

Target Roles: Cloud Data Platform Engineer | AWS Platform Engineer | DataOps Engineer

- **Platform Modernization & Version Remediation:** Spearheaded organization-wide EMR runtime and Python module upgrades across DEV, PRE, and PROD environments, remediating 100% of end-of-support clusters within strict SLA windows and completely eliminating legacy runtime security vulnerabilities.
- **Cluster Reliability & Observability:** Implemented automated AWS Health Dashboard tracking and SNS alerting integrated with EC2/EMR IAM service roles, driving down incident response times and establishing standardized operational recovery runbooks.
- **Troubleshooting & Workload Optimization:** Onboarded multi-tenant engineering teams to EMR Advisor and resolved critical Lake Formation Iceberg fine-grained access control (FGAC) and YARN startup failures, restoring operational uptime for high-throughput production pipelines.

2. Data Governance, Security & Access Automation

Target Roles: Data Governance Specialist | Cloud Security & Identity Engineer | Data Platform Operations Specialist

- **Automated Access Management:** Delivered a web-based self-service Lake Formation Access Automation workflow, cutting manual provisioning lead time from multiple days to under an hour and eliminating repetitive ticket overhead.
- **Fine-Grained Access Control (FGAC):** Architected and deployed AWS Lake Formation Tag-Based Access Control (TBAC/LF-Tags) across multi-account enterprise environments, standardizing dataset-level permissions and ensuring strict regulatory compliance.
- **Cross-Account Identity & Storage Security:** Formulated enterprise IAM role governance, ARN-mapping policies, and S3 bucket security controls, enabling zero-

trust cross-account data ingestion without compromising principle-of-least-privilege boundaries.

3. Data Producer/Consumer Onboarding & CI/CD Automation

Target Roles: Platform DevOps Engineer | Data Platform Operations Lead | Technical Onboarding Engineer

- **Zero-Touch Team Onboarding:** Built automated Terraform-driven repository provisioning for data producer and consumer teams, reducing onboarding turnaround time by ~70% and standardizing user group entitlements across business units.
- **Secure CI/CD Engineering:** Optimized Terraform deployment pipelines by resolving multi-account AWS OIDC AssumeRole authentication issues and migrating repository security scanning from `tfsec` to `trivy`, achieving 100% static-analysis pipeline compliance.
- **Analytics & Data Science Enablement:** Authored end-to-end operational runbooks and designed streamlined onboarding workflows for Amazon SageMaker and analytics teams, shortening time-to-first-query for downstream data scientists.

4. Cloud Data Warehousing & Modern Analytics (Snowflake & Athena)

Target Roles: Snowflake / Cloud Data Engineer | Data Warehouse Administrator | BI Platform Engineer

- **Enterprise Snowflake Provisioning:** Configured and managed Snowflake Data Exchange infrastructure, custom role-based access controls (RBAC), and multi-environment (PRE/PROD) promotion workflows supporting high-concurrency analytical workloads.
- **Automated Metadata & Asset Tagging:** Implemented automated, scheduled historical metadata tagging across Snowflake schemas and tables, ensuring 100% compliance with corporate data audit and lineage standards.
- **Query Performance & Cost Governance:** Enforced AWS Athena workgroup data usage limits and fine-tuned partitioned table access patterns, preventing runaway ad-hoc query costs while eliminating execution bottlenecks for BI consumers.

5. Enterprise Data Migration & Solution Architecture

Target Roles: Data Solutions Architect | Technical Product Owner / Business Analyst | Data Migration Lead

- **Common Data Architecture Design:** Produced comprehensive enterprise architecture specifications for batch, streaming, and outbound SFTP patterns within the Common Data Framework (CDF), accelerating cross-functional engineering alignment.
- **Migration Strategy & Tooling:** Authored the enterprise Data Science Migration Toolkit ("How-To" blueprint) and established a multi-tier prioritization matrix for migrating legacy Data Center of Excellence use cases to the cloud.
- **Cloud Readiness & Decommission Audits:** Conducted rigorous application and model inventory assessments across Risk, Compliance, and COO business units, establishing technical roadmaps for legacy system decommission and seamless cloud transition.